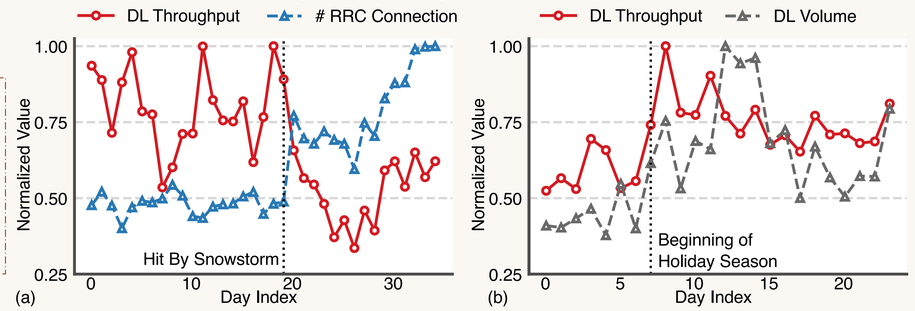
Predicting the Performance of Cellular Networks: A Latent-resilient Approach

Kartik Patel[1], Changhan Ge[1], Ajay Mahimkar[2], Sanjay Shakkottai[1], Yusef Shaqalle[2]
[1]The University of Texas at Austin, Austin, TX 78705, USA
[2]AT&T Labs – Research, USA

Motivation & Contribution

Motivation: Predicting the performance impacts of configuration changes in current cellular network is **non-trivial**, due to

Examples of the Impact of Latent Variables



Contributions:

- Mitigated the effects of latents by considering intermediates, called **Mobility, Access, and Traffic (MAT) metrics**
- Proposed the **C**onfiguration **I**mpact **P**rediction **A**nalysis **T**oolkit (**CIPAT**) driven by a large real-world dataset from live LTE and 5G networks.

Two-stage Framework and CIPAT Design

Stage 1: Configuration -> MAT metrics

- Use CODEC [1] to measure the conditional dependency between features (attributes or configurations) and target variables (MAT metrics or KPIs).

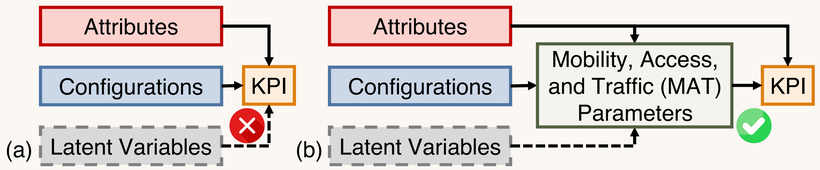
$$T(x, P|Q) = \frac{\int \mathbb{E}(\mathrm{Var}(\mathbb{P}(x \geq t|P,Q)|Q))d\mu(t)}{\int \mathbb{E}(\mathrm{Var}(\mathbf{1}\{x \geq t\}|Q))d\mu(t)}$$

- Use FOCI [1] to rank and identify the key configuration parameters for each MAT metric.

Stage 2: Predicting MAT metrics -> KPI

- MAT → KPIs is non-linear. We designed a DNN to map MAT to KPIs.

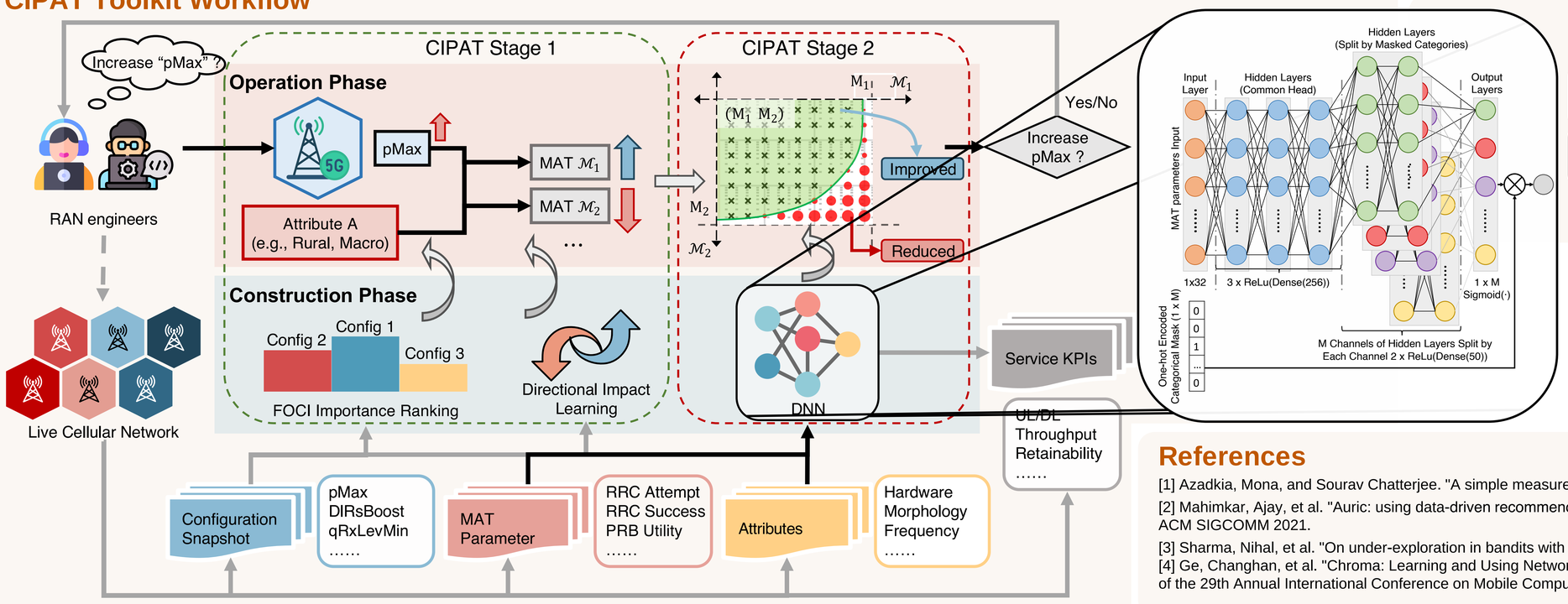
Dependency Graph (a) One-stage. (b) Two-stage



MAT metrics better predict KPIs compared to configurations

Network	KPI (x)	Configurations (P)	MAT (P)
LTE	DL Throughput	0.2323	**0.6176**
	VoLTE Throughput	0.2999	**0.4093**
5G	DL Throughput	0.0959	**0.6641**
	NR Retainability	0.0512	**0.4613**

CIPAT Toolkit Workflow



Evaluation

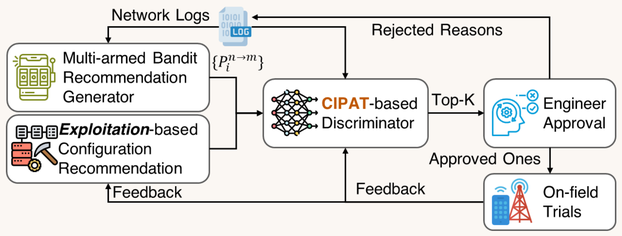
MSE of models mapping MAT metrics to KPIs

Model	Mask/Feature Attributes (M/F)					MSE	wrt Var – 0.015	
	MIMO Mode	Freq	eNB Type	Hard-ware	Soft-ware		Error	Improv-ement
One-stage	M	F	F	F	F	0.00638	42.54%	N/A
Two-stage (DNN)	F	F	F	F	F	0.00185	12.34%	+30.20%
	M	F	F	F	F	0.00197	13.10%	+29.44%
	M	M	F	F	F	0.00218	14.53%	+28.10%
	M	M	M	F	F	0.00204	13.62%	+28.92%
	M	M	M	M	M	0.00197	13.16%	+29.38%
Common Mask Attributes					Market, Morphology, Tower Height			
Common Feature Attributes					Weekday, Bandwidth			

Performance of CIPAT

Market & Attributes	Accuracy	Efficacy	
		All Changes	Novel Changes
(A) Densely Populated Suburban Plain terrain	0.74	0.86	0.93
(B) Densely populated metropolitan Downtown, Plain terrain	0.85	0.82	0.95
(D) Densely populated urban and sparsely populated mountains and deserts	0.7	0.78	0.76
(E) Sparsely populated regions with mountains, plateaus, and deserts	0.78	0.78	0.81

Application



Acknowledgement

The authors greatly appreciate the fruitful comments from MobiCom 24 anonymous reviewers. This work is supported by NSF grants CNS-2107037 and CNS-2112471. Changhan Ge sincerely appreciate the support from Prof. Lili Qiu at UT Austin. The work does not raise any ethical concerns.

References

[1] Azadkia, Mona, and Sourav Chatterjee. "A simple measure of conditional dependence." The Annals of Statistics 49.6 (2021): 3070-3102
[2] Mahimkar, Ajay, et al. "Auric: using data-driven recommendation to automatically generate cellular configuration." Proceedings of the ACM SIGCOMM 2021.
[3] Sharma, Nihal, et al. "On under-exploration in bandits with mean bounds from confounded data." arXiv preprint arXiv:2002.08405 (2020).
[4] Ge, Changhan, et al. "Chroma: Learning and Using Network Contexts to Reinforce Performance Improving Configurations." Proceedings of the 29th Annual International Conference on Mobile Computing and Networking. 2023.